The tangent space of M_{Higgs} at (E,ϕ)

Consider the complex

$$C^\bullet : \quad 0 \longrightarrow \mathcal{E}nd\,E \xrightarrow{[\cdot,\phi]} \mathcal{E}nd\,E \otimes K \longrightarrow 0 .$$

We claim that

$$T_{(E,\phi)} M_{Higgs} = \mathbb{H}^1(C^\bullet).$$

Let $\mathcal{U} = \{U_i = \operatorname{Spec} A_i\}$ be an affine. Let $\mathcal{E}nd\,E|_{U_i} = \tilde{M}_i$, $\mathcal{E}nd\,E\otimes K|_{U_i} = \tilde{R}_i$. We get an acyclic resolution

$$\begin{tikzcd}
\vdots & \vdots & \vdots & \vdots \\
0 \arrow[u] \arrow[r] & \bigoplus \tilde{M}_{ij} \arrow[u] \arrow[r, "{[\cdot,\phi]}"] & \bigoplus \tilde{M}_{ij}\otimes R_{ij} \arrow[u] \arrow[r] & 0 \arrow[u] \\
0 \arrow[u] \arrow[r] & \bigoplus \tilde{M}_i \arrow[u] \arrow[r, "{[\cdot,\phi]}"] & \bigoplus \tilde{M}_i \otimes R_i \arrow[u] \arrow[r] & 0 \arrow[u] \\
0 \arrow[u] \arrow[r] & \mathcal{E}nd\,E \arrow[u] \arrow[r, "{[\cdot,\phi]}"] & \mathcal{E}nd\,E\otimes K \arrow[u] \arrow[r] & 0. \arrow[u]
\end{tikzcd}$$

Hence, $\mathbb{H}^1(C^\bullet)$ consists of pairs $((s_{ij}),(t_k))$ s.t.

$(s_{ij}) \in H^1(\mathcal{E}nd\,E)$ is a Čech cocycle,

$\alpha_i : M_i \longrightarrow M_i \otimes R_i$ satisfies

$$(\alpha_i - \alpha_j)|_{U_{ij}} = [s_{ij}, \phi].$$

Let now $\mathbb{C}[\varepsilon] = \mathbb{C}[z]/z^2$. Consider

$$\begin{tikzcd}
X \arrow[r] \arrow[d] & X \times \operatorname{Spec}\mathbb{C}[\varepsilon] = X_\varepsilon \arrow[d] \\
\operatorname{Spec}\mathbb{C} \arrow[r] & \operatorname{Spec}\mathbb{C}[\varepsilon]
\end{tikzcd}$$

via $\mathbb{C}[\varepsilon] \xrightarrow{\varepsilon \mapsto 0} \mathbb{C}$

The tangent space $T_{(E,\phi)} M$ consists of all Higgs bundles $(E_\varepsilon, \phi_\varepsilon)$ on X_ε s.t. the restriction to $X \times \operatorname{Spec}\mathbb{C} = X$ is (E,ϕ).
Let $U_i[\varepsilon] = U_i \times \operatorname{Spec}\mathbb{C}[\varepsilon]$. Then any such $(E_\varepsilon,\phi_\varepsilon)$ is trivial on $U_i[\varepsilon]$ and has transition functions $f_{ij} \in \operatorname{End}(M_{ij}[\varepsilon])$. Since f_{ij} is 1 at $\varepsilon = 0$, $f_{ij} = 1 + \varepsilon s_{ij}$, where $s_{ij} \in \operatorname{End}(M_{ij})$. By the cocycle condition,

probably the cocycle for E, not 1; replace it everywhere

$$(s_{ij}) \in H^1(\mathcal{E}nd\,E).$$

Conversely, any element in $H^1(\mathcal{E}nd\,E)$ determines a deformation of E.
Any $\phi_\varepsilon : E_\varepsilon \longrightarrow E_\varepsilon \otimes K$ is given by maps $\phi^i_\varepsilon : M_i[\varepsilon] \longrightarrow M_i \otimes R_i[\varepsilon]$ s.t.

$$\begin{tikzcd}
M_{ij}[\varepsilon] \arrow[r, "\phi^i_\varepsilon"] \arrow[d, "1+\varepsilon s_{ij}"] \arrow[dr, phantom, "\circlearrowleft"] & M_{ij}\otimes R_{ij}[\varepsilon] \arrow[d, "(1+\varepsilon s_{ij})\otimes 1"] \\
M_{ij}[\varepsilon] \arrow[r, "\phi^j_\varepsilon"] & M_{ij}\otimes R_{ij}[\varepsilon]
\end{tikzcd}$$

Since $\phi_\varepsilon = \phi$ at $\varepsilon = 0$, we have $\phi^i_\varepsilon = \phi^i + \varepsilon\alpha_i$. Plugging this in and restricting to U_{ij}, we get

$$\begin{aligned}
(1+\varepsilon s_{ij})\otimes 1 \circ (\phi^i + \varepsilon\alpha_i) &= (\phi^i + \varepsilon s_{ij}\phi^i + \varepsilon\alpha_i) \\
= (\phi^j + \varepsilon\alpha_j)\circ(1+\varepsilon s_{ij}) &= \phi^j + \varepsilon\alpha_j + \varepsilon\phi^j s_{ij} .
\end{aligned}$$

we did not twist M_{ij} or R_{ij}, ie replace 1 by cocycles

Hence,

$$(s_{ij}\phi^i - \phi^j s_{ij})|_{U_{ij}} + (\alpha_i - \alpha_j)|_{U_{ij}} = 0.$$

Since $\phi^i|_{U_{ij}} = \phi^j|_{U_{ij}}$, we get

$$[s_{ij}, \phi] = (\alpha_j - \alpha_i)|_{U_{ij}} .$$

Hence, the pair $(s_{ij}, \alpha_i) \in \mathbb{H}^1(C^\bullet)$. The converse is clear.

By Serre duality, if

$$F^\bullet : \quad 0 \longrightarrow F^0 \longrightarrow F^1 \longrightarrow \cdots \longrightarrow F^n \longrightarrow 0$$

is a complex of loc. free sheaves then

$$\mathbb{H}^i(F^\bullet)^\vee = \mathbb{H}^{n-i+1}(\check{F}^\bullet),$$

where $\check{F}^j = (F^{n-j})^\vee \otimes K$. Hence, $\mathbb{H}^1(C^\bullet)^\vee$ of the complex

$$C^\bullet : \quad 0 \longrightarrow \mathcal{E}nd\,E \xrightarrow{[\cdot,\phi]} \mathcal{E}nd\,E \otimes K \longrightarrow 0 \qquad \mathbb{H}^1(F^\bullet)^\vee = \mathbb{H}^1(\check{F}^\bullet)$$

is $\mathbb{H}^1$ of the complex

$$C_\bullet : \quad 0 \longrightarrow (\mathcal{E}nd\,E\otimes K)^\vee \otimes K \xrightarrow{[\cdot,\phi]^T \otimes \mathrm{id}} \mathcal{E}nd\,E^\vee \otimes K \longrightarrow 0$$

ie

$$C_\bullet : \quad 0 \longrightarrow \mathcal{E}nd\,E \xrightarrow{[\phi,\cdot]} \mathcal{E}nd\,E \otimes K \longrightarrow 0 .$$